STAT201 · STATISTICS · LECTURE 9

Estimation and Confidence Intervals

How to report, instead of a single number, a range of plausible values of a parameter and honestly measure uncertainty

Syllabus topic 9 · week 9 · 4 academic hours

LECTURE GOALS

- distinguish point and interval estimates and explain the properties of a good estimator;
- construct confidence intervals for a mean (with known and unknown σ) and for a proportion;
- use the t-distribution, find the degrees of freedom and critical values;
- interpret the confidence level correctly and avoid typical wording errors;
- assess how the width of an interval depends on n, σ and the confidence level; determine the required sample size.

1 Point and interval estimates

A *point estimate* is a single number computed from a sample: $\bar{x}$ for μ, $\hat{p}$ for p, s^2 for σ^2. A good estimator is:

- *unbiased* — its average over all samples equals the parameter (Lecture 7);
- *consistent* — it approaches the parameter as n grows;
- *efficient* — it has the smallest variance among unbiased estimators.

But a single number says nothing about precision. A sample mean of 35.75 may be close to μ or far from it — that depends on the SE. A *confidence interval* (CI) gives a range of plausible values of the parameter together with a measure of the reliability of the method.

CONFIDENCE INTERVAL

An interval with confidence level $1-\alpha$ is built from a sample so that in the long run (under repeated sampling) the share of intervals covering the true parameter equals $1-\alpha$. The general form:

$$\text{estimate} \;\pm\; \underbrace{\text{critical value} \times \text{SE}}_{\text{margin of error } E}. \tag{1.1}$$

2 CI for the mean when σ is known

If σ is known (rare, but a convenient starting point) and either the population is normal or $n \geq 30$, then by the CLT

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad z_{0.05} = 1.645\ (90\%), \quad z_{0.025} = 1.960\ (95\%), \quad z_{0.005} = 2.576\ (99\%). \tag{2.1}$$

EXAMPLE 1. APPLICATION PROCESSING TIME

A sample of $n = 36$ applications, $\bar{x} = 12.4$ days, $\sigma = 4$. SE $= 4/6 = 0.667$. 95% CI: $12.4 \pm 1.96 \cdot 0.667 = 12.4 \pm 1.31$, i.e. $(11.09;\ 13.71)$. 99% CI: $12.4 \pm 2.576 \cdot 0.667 = (10.68;\ 14.12)$ — wider.

3 How to understand the confidence level correctly

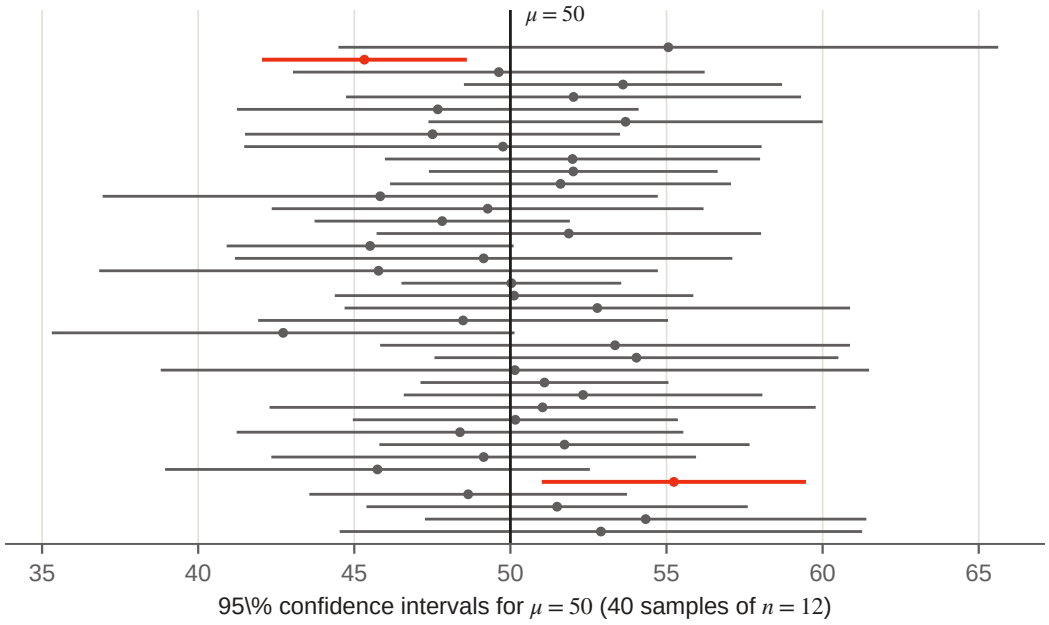


Fig. 1. 40 independent samples of 12 observations from $N(50, 10^2)$; each gives its own 95% interval. Two of the forty (red) did not cover $\mu = 50$ — exactly what 95% confidence allows.

The parameter μ is fixed, the interval's endpoints are random. "95% confidence" is a property of the *method*: 95% of the intervals built this way cover μ. For a specific computed interval $(11.09; 13.71)$ it is incorrect to say "with probability 95% μ lies in this interval" (it either contains μ or it does not), but it is acceptable to say: "we are 95% confident that the interval covers μ" or "values of μ from 11.1 to 13.7 are consistent with the data".

COMMON MISTAKE

Wrong: "95% of the observations lie in the confidence interval". A CI is for the *mean*, not for individual observations; the limits for 95% of observations (a prediction interval) are much wider.

4 Unknown σ: the t-distribution

Usually σ is unknown and is replaced by the sample s. But s is itself random, especially for small n. The statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \tag{4.1}$$

has not a normal but a *Student distribution* with $\nu = n - 1$ degrees of freedom (if the population is normal). It is symmetric like the normal but has "heavier tails"; as ν grows it approaches $N(0, 1)$.

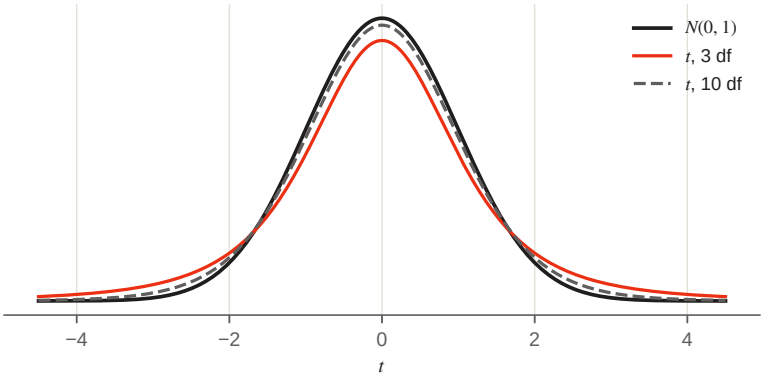


Fig. 2. The Student distribution with few degrees of freedom is wider than the normal; with 10 degrees of freedom it is already almost indistinguishable.

CI for the mean when σ is unknown

$$\bar{x} \pm t_{\alpha/2,\,n-1}\,\frac{s}{\sqrt{n}}.$$

Critical values $t_{0.025,\nu}$ (for 95%): $\nu = 4$: 2.776; 9: 2.262; 11: 2.201; 14: 2.145; 24: 2.064; 29: 2.045; 59: 2.001; 99: 1.984; ∞: 1.960 (the full table is in the appendix).

EXAMPLE 2. COFFEE SHOPS: MONTHLY REVENUE

From Lecture 1: 12 branches of a coffee-shop chain, $\bar{x} = 35.75$, $s = 20.33$ (monthly revenue of a branch, million UZS). SE $= 20.33/\sqrt{12} = 5.869$, $t_{0.025,11} = 2.201$, $E = 2.201 \cdot 5.869 = 12.92$. 95% CI: $35.75 \pm 12.92 = (22.83;\ 48.67)$. 99% CI ($t_{0.005,11} = 3.106$): $35.75 \pm 18.23 = (17.52;\ 53.98)$. If $z = 1.96$ had been used by mistake, the margin of error would have been 11.50 instead of 12.92 — the interval would have been too narrow and would "promise" more than there really is.

REMEMBER

Conditions: (1) the sample is random, observations are independent; (2) the population is approximately normal *or* n is large enough ($\gtrsim 30$). For $n < 30$ with clear skewness or outliers the t-interval is unreliable: in the coffee-shop example the outlier (branch 12) strongly affects $\bar{x}$ and s, and the result should be supplemented by an interval without it.

5 What determines the width of an interval

The margin of error $E = t \cdot s/\sqrt{n}$ decreases when: n grows (by a factor of $\sqrt{n}$ — four times as much data for an interval half as wide); the spread s decreases; the confidence level decreases (90% is narrower, 99% wider). The choice of confidence level is a compromise between certainty and precision; by default 95%.

6 CI for a proportion

CI for a proportion

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad (\text{for } n\hat{p} \geq 10,\ n(1-\hat{p}) \geq 10).$$

EXAMPLE 3. SHARE OF SATISFIED CUSTOMERS

Of 400 respondents 240 are satisfied: $\hat{p} = 0.60$, SE $= \sqrt{0.6 \cdot 0.4/400} = 0.0245$. 95% CI: $0.60 \pm 1.96 \cdot 0.0245 = 0.60 \pm 0.048 = (0.552;\ 0.648)$. 90% CI: $(0.560;\ 0.640)$.

For small samples and extreme proportions the simple interval is unreliable. For $n = 20$, $x = 2$ ($\hat{p} = 0.10$) it gives $(-0.031;\ 0.231)$ — with a negative lower limit; the Wilson interval $(0.028;\ 0.301)$ is more correct. Rule: if the conditions $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ are not met, use the Wilson or an exact interval.

7 Sample size for a given precision

$$n \geq \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2, \qquad n \geq \frac{z_{\alpha/2}^2\,\hat{p}(1-\hat{p})}{E^2} \ \text{(in the worst case } \hat{p} = 0.5). \tag{7.1}$$

For a mean with $\sigma \approx 20$, $E = 3$ and 99% confidence: $n \geq (2.576 \cdot 20/3)^2 = 294.9$, i.e. 295. For a proportion with $E = 0.03$ and a preliminary estimate $p \approx 0.6$: $n \geq 1.96^2 \cdot 0.24/0.0009 = 1024.4$, i.e. 1025; without a preliminary estimate (worst case) 1068. The value of σ is taken from a pilot sample or past data; with a t-interval the final n is slightly larger.

8 CI for a variance (for information)

If the population is normal, then $(n-1)s^2/\sigma^2$ has a χ^2 distribution with $n-1$ degrees of freedom, and

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \ \leq\ \sigma^2 \ \leq\ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}. \tag{8.1}$$

For the coffee shops: $s^2 = 413.3$, $\chi^2_{0.025,11} = 21.92$, $\chi^2_{0.975,11} = 3.816$: $\sigma^2 \in (207.4;\ 1191.5)$, i.e. $\sigma \in (14.4;\ 34.5)$. The interval is very wide: with $n = 12$ the spread is estimated poorly. This interval is sensitive to departures from normality.

9 Key terms

Term	*In Russian*	Meaning
Point estimate	*Точечная оценка*	a single number from a sample
Confidence interval (CI)	*Доверительный интервал*	estimate $\pm$ critical value $\times$ SE
Confidence level	*Уровень доверия*	$1-\alpha$
Margin of error	*Предельная ошибка*	E — half the width of the CI
Degrees of freedom	*Степени свободы*	$\nu = n-1$ for a mean
Student's t distribution	*Распределение Стьюдента*	heavy tails, $\to N(0,1)$

Formulas of Lecture 9

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad \bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}}, \qquad \hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}};$$

$$n \geq \left(\frac{z\sigma}{E}\right)^2, \qquad n \geq \frac{z^2\hat{p}(1-\hat{p})}{E^2}.$$

10 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] The mean score of a sample of 49 students is 82, $\sigma = 14$. Construct a 95% CI for the mean.

Problem 2 [A] Find from the table the critical value t for: (a) a 95% CI, $n = 10$; (b) a 99% CI, $n = 25$; (c) a 90% CI, $n = 16$.

Problem 3 [A] A sample of 25 customers: $\bar{x} = 82$ thousand UZS, $s = 10$. Construct a 95% CI for the average bill. Why is t used here rather than z?

Problem 4 [B] The cost of 16 randomly selected orders: $\bar{x} = 250$ thousand, $s = 40$ thousand. Construct a 99% CI for the average order cost and state what happens to the width when moving to 95%.

Problem 5 [B] Of 500 transactions checked, 85 are flagged as suspicious. Construct 95% and 90% CIs for the share of suspicious transactions.

Problem 6 [B] Delivery times of 8 orders (minutes): $18, 22, 25, 21, 30, 24, 27, 23$.

(a) Find $\bar{x}$ and s.
(b) Construct a 95% CI for the mean delivery time.
(c) What assumption is needed, and how can it be checked on such data?

Problem 7 [B] To estimate the average response time ($\sigma \approx 20$ s) a margin of error $E = 3$ s is required. Find n for 95% and 99% confidence. Then find n for estimating the share of satisfied customers with $E = 0.03$ at 95% confidence if (a) p is unknown; (b) preliminarily $p \approx 0.6$.

Problem 8 [B] An analyst writes: "With probability 95% the average processing time lies in the interval $(22.8;\ 48.7)$" and "95% of branches have revenue in this interval". Find and correct the errors in both statements.

Problem 9 [B] How does the width of a CI for a mean (σ known) change if (a) n grows fourfold; (b) the confidence level changes from 95% to 99%; (c) σ decreases by 20%?

Problem 10 [B] In a sample of 80 borrowers, 6 were overdue. Check the conditions, construct the simple 95% CI for the proportion and the Wilson interval. Compare and comment.

Problem 11 [C] Derive the CI for μ with known σ from the condition $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$, where $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$.

Problem 12 [C] For a sample of $n = 2$ observations $t_{0.025,1} = 12.706$. Explain why a 95% CI with $n = 2$ is almost useless, while for $n = 100$ ($t = 1.984$) it practically coincides with the z-interval. What role does the estimate s play here?